

EXPT NO:7	IMPLEMENT ADVANCED CHART TYPES USING TABLEAU AND POWER BI
DATE: 14.08.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is data integration in the context of Business Intelligence (BI)?

Data integration in Business Intelligence means combining data from different sources into a common system so that it can be analyzed together. The sources can be CSV files, Excel files, databases, or other systems. For example, in this experiment, sales information is taken from a CSV file and regional target information is taken from an Excel file. By integrating both datasets, we can compare actual sales with the target and get a better understanding of business performance.

2. Explain the difference between a join and a data blend in Tableau.

- A **join** combines two tables using a common field such as Region. The tables are combined at the data source level, so the resulting data can be used for analysis.
- A **data blend** is used when the data comes from different data sources. Tableau connects the sources using a common field and combines the information while creating the visualization. In simple words, a join combines tables, while a blend combines data from separate sources.

3. What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?

A **relationship** in Power BI is used to connect two tables through a common column. It allows data from both tables to work together in reports and visuals. **Cardinality** describes how the records in one table are related to records in another table. The common types are one-to-one, one-to-many, and many-to-many. In this experiment, many sales records can belong to one region, so a many-to-one relationship is suitable.

4. Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?

Data quality is important because incorrect or inconsistent data can produce wrong results in the dashboard. When integrating different sources, column names, values, and data types should be checked properly. For example, if one dataset contains "N. America" and the other contains "North America", they may not be matched correctly. Similarly, sales should be stored as numbers and order dates should be stored as dates. Cleaning the data before integration helps to get accurate results.

5. Name two common challenges encountered during data integration.

Two common challenges are:

- Different naming formats or mismatched keys between datasets.
- Missing values or incorrect data types.

IN-LAB EXERCISE:

OBJECTIVE:

To connect, combine, and model data from multiple sources in Tableau and Power BI and create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

1. Data Collection:

- Import two datasets:
 - Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
 - Dataset 2: Regional Sales Targets (Excel) → Region, Sales Target, Target Year

2. Data Preparation and Transformation:

- Clean datasets (e.g., standardize "N. America" → "North America").
- Ensure correct data types (numeric for Sales, date for Order Date).
- Handle missing values appropriately.

3. Data Integration and Modeling:

- **Tableau:** Create a join/relationship on the Region field in the Data Source pane.
- **Power BI:** Use the Model view to create a relationship on Region; define cardinality (e.g., many-to-one).

4. Create Calculated Fields for Comparison:

- Target Achievement % = $(\text{SUM}(\text{Sales}) / \text{SUM}(\text{Sales Target})) * 100$
- Sales vs. Target Variance = $\text{SUM}(\text{Sales}) - \text{SUM}(\text{Sales Target})$

5. Design a Comprehensive Dashboard:

- Combine visuals from both sources.
- Add KPI cards: Total Sales, Overall Target Achievement %.
- Use gauge/bullet charts for region-wise target performance.
- Combined bar chart: Actual Sales vs. Target by Region.
- Map visualization: Sales vs. Target Variance by region.

6. Validate Data Integrity:

- Cross-check joined data for accuracy.
- Ensure dashboard filters correctly update visuals.

7. Publishing and Sharing:

- **Tableau:** Publish to Tableau Public or Server.
- **Power BI:** Publish to Power BI Service and share.

Experiment Datasets:

- **Dataset 1:** Retail Sales Data (CSV) → Order ID, Order Date, Region, Sales
- **Dataset 2:** Regional Sales Targets (Excel) → Region, Sales Target, Target Year

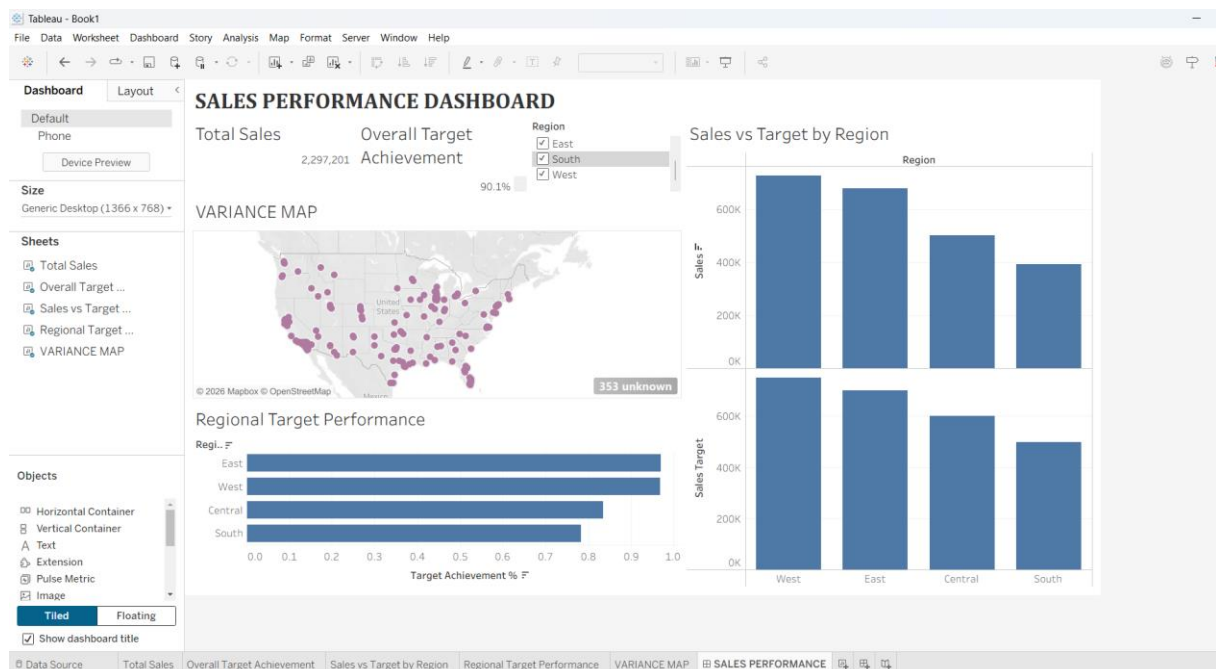
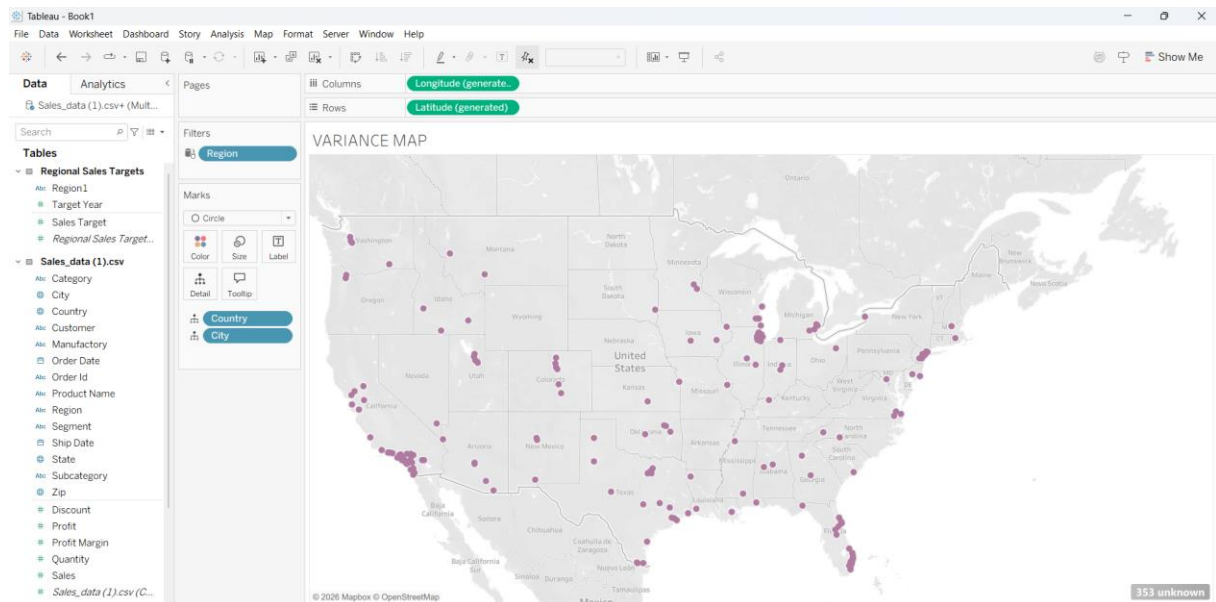
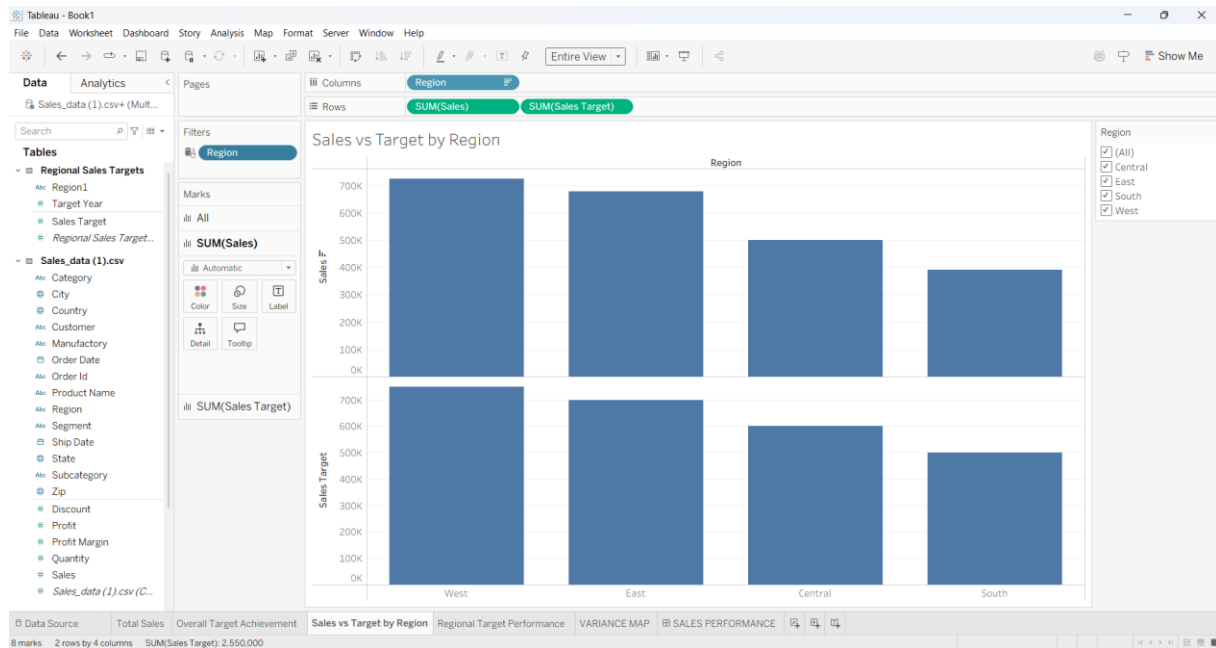
TABLEAU STEPS :

The screenshot shows the Tableau Desktop interface with the 'Sales_data (1).csv+ (Multiple Connections)' data source. The 'Regional Sales Targets' table is joined to the 'Sales_data (1).csv' table. The join configuration is set to 'Region1' with a cardinality of 'Many' and a referential integrity of 'Some records match'. The 'Performance Options' section is expanded, showing settings for cardinality and referential integrity.

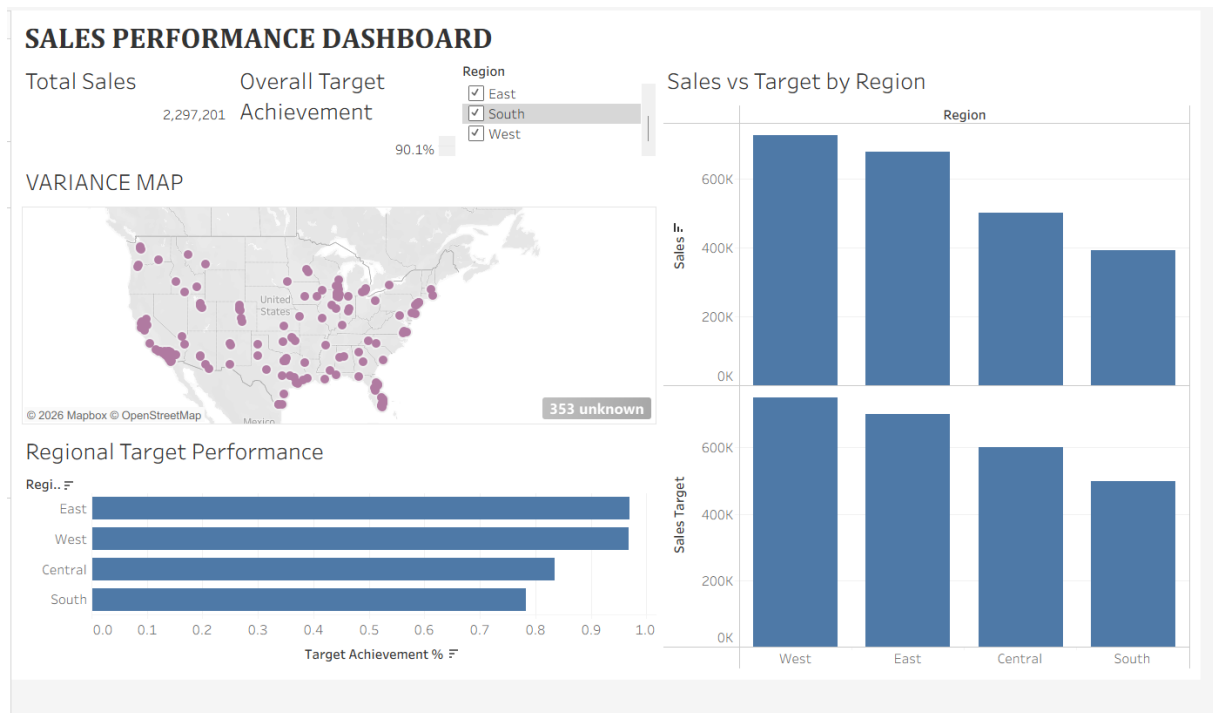
Region1	Sales Target	Target Year
Central	600,000	2024
East	700,000	2024
South	500,000	2024
West	750,000	2024

The screenshot shows the final Tableau dashboard view. The 'Total Sales' card displays a value of 2,297,201. The 'Filters' shelf contains the 'Region' field. The 'Marks' shelf contains the 'SUM(Sales)' field. The 'Columns' shelf is empty. The 'Rows' shelf is empty. The 'Tables' pane on the left shows the 'Regional Sales Targets' and 'Sales_data (1).csv' tables.

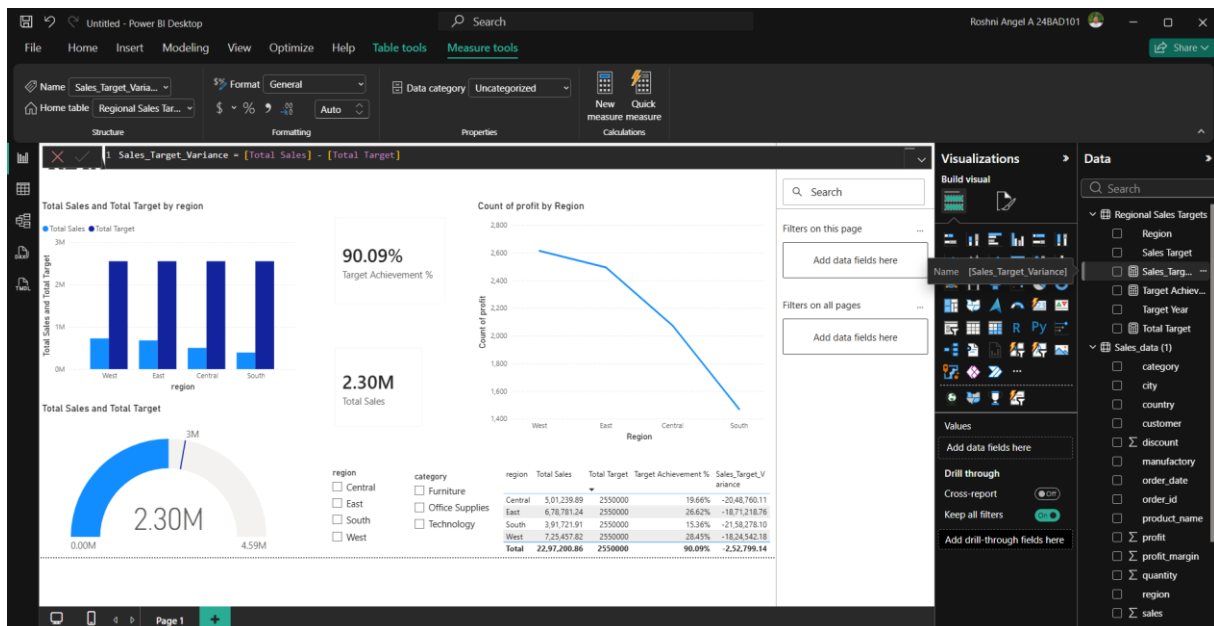
Region	Total Sales
Central	600,000
East	700,000
South	500,000
West	750,000

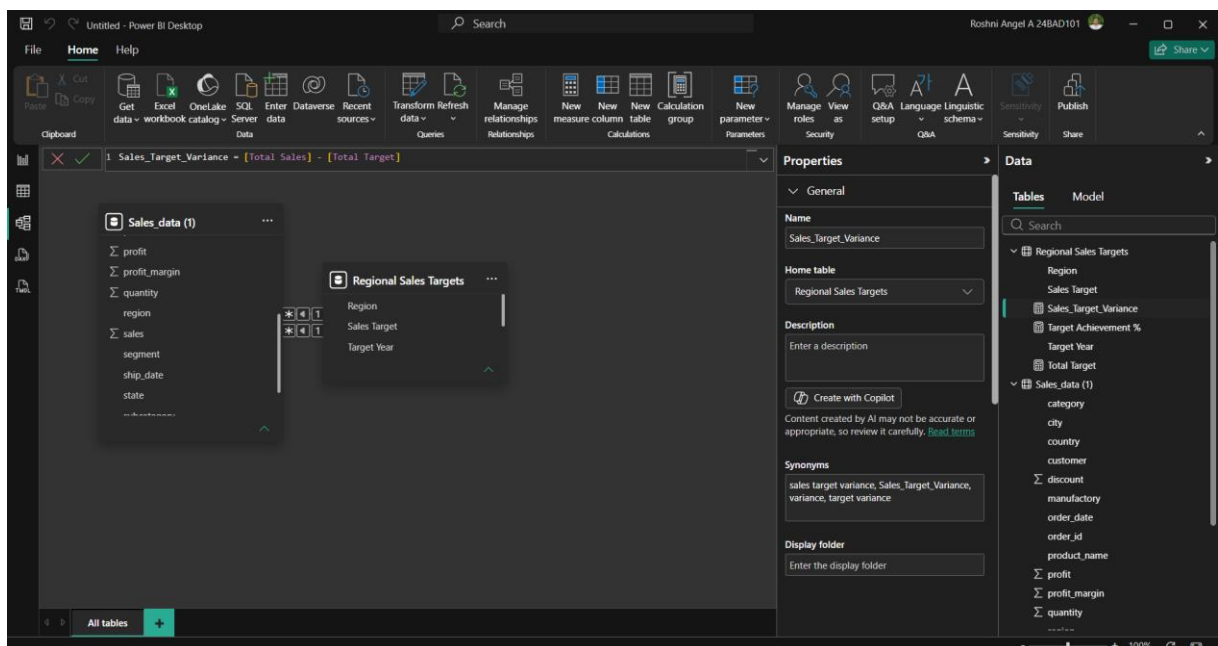
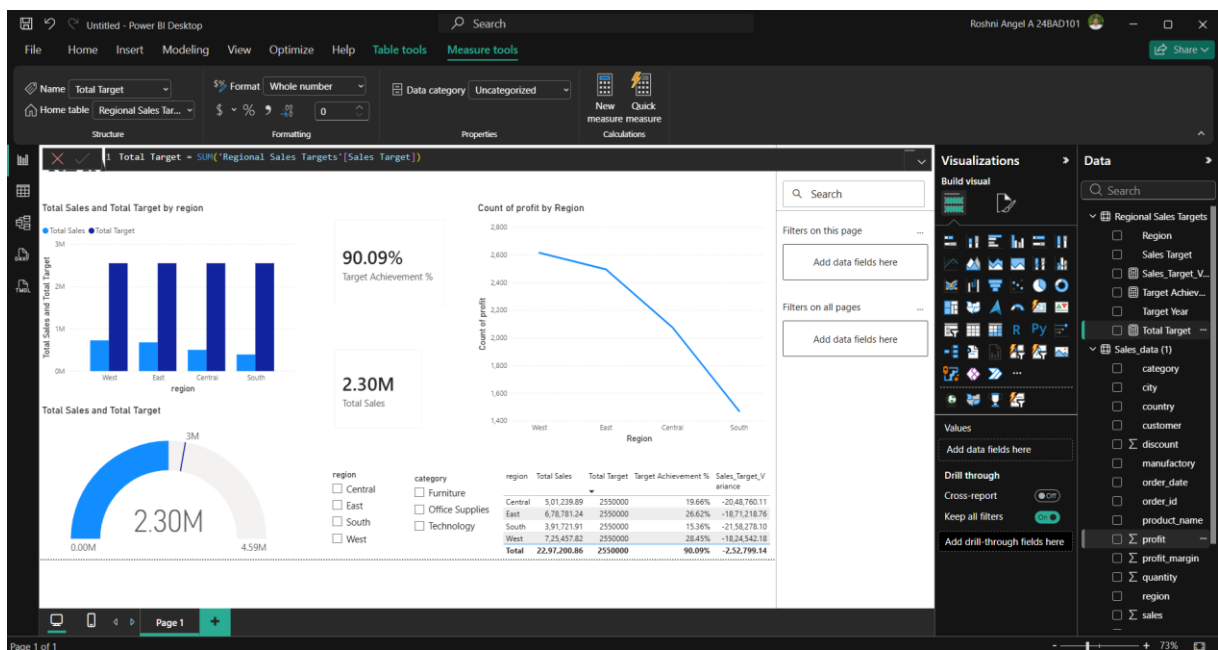
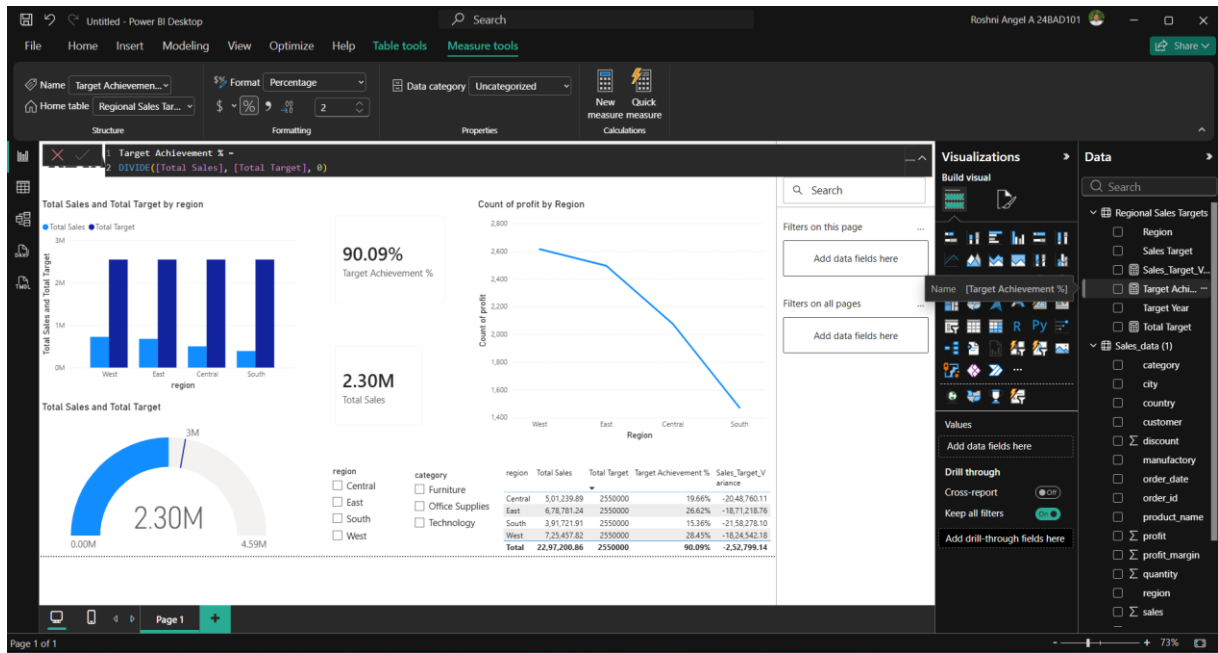


FINAL DASHBOARD :



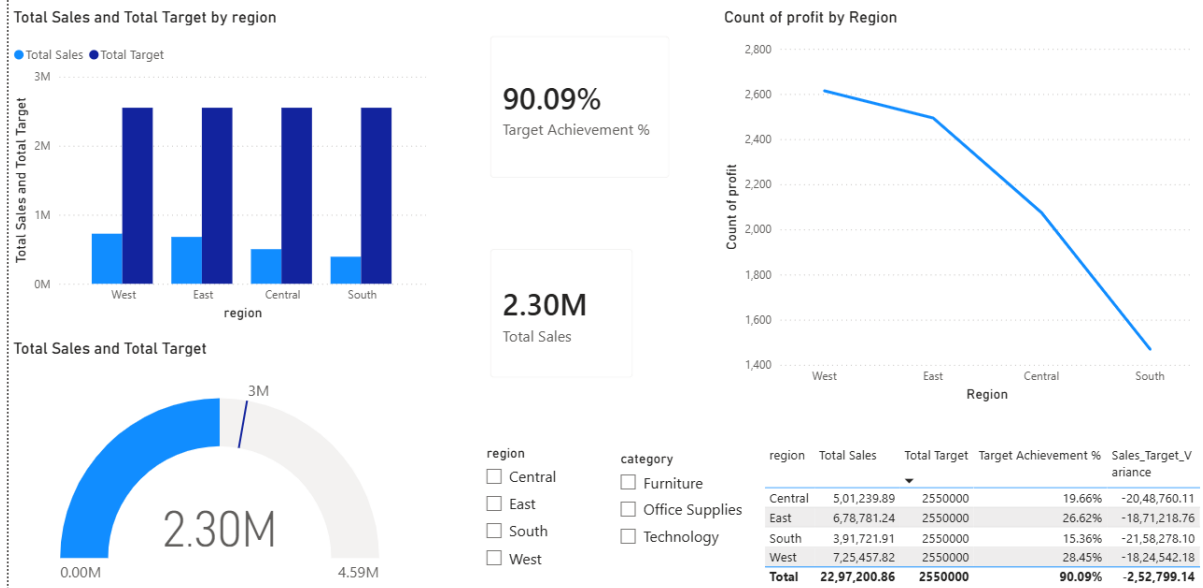
POWER BI STEPS :





FINAL DASHBOARD :

RETAIL SALES PERFORMANCE



POWER BI LINK :

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POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?

- Using only sales data tells us how much each region sold, but it does not tell us whether the sales were good enough compared to the expected target. After integrating the regional sales targets, we could compare actual sales with the target for each region.
- This helped us identify regions that achieved or exceeded their targets and regions that were below their targets. Therefore, the integrated data provided a better view of the overall business performance.

2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.

- The two datasets were connected using the **Region** field because it was available in both datasets. In Power BI, a **many-to-one relationship** was appropriate because the sales dataset can contain many sales records for the same region, while the target dataset contains the target value for a region.
- In Tableau, the datasets can be connected using a relationship or join based on the Region field. This allows actual sales and target values to be compared in the same dashboard.

3. What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?

- One challenge was ensuring that the region names were exactly the same in both datasets. For example, “**N. America**” needed to be changed to “**North America**” so that the records could match correctly.
- Another thing I checked was the data type of each column. Sales values had to be numeric, while Order Date had to be recognized as a date. I also checked for missing values before creating the relationship.

4. How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?

- Variance calculations show the difference between actual sales and the target sales. If the variance is positive, it means the region has performed better than the target. If it is negative, the region has not reached its target.
- This gives the manager a quick idea of which regions are doing well and which ones need more attention. Based on this information, the manager can plan new sales strategies, improve marketing, or provide additional support to weaker regions.

5. What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?

- I could integrate **marketing campaign data** with the existing sales and target data. This could help us understand whether a particular marketing campaign resulted in increased sales in a region.
- We could also integrate **customer demographic data** such as age group, location, or customer type. This could help identify which types of customers are contributing most to sales and help the business plan better marketing strategies.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in Tableau and Power BI. Relationships between the sources created a unified data model. The dashboard visualized sales performance against targets, enabling variance analysis and providing insights into regional performance not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		